

Generation Interconnection System Impact Study Report

For Queue Project Q19-1320

Middlesex 138 kV
November 20, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1320. The Interconnection Customer proposes a 260.3 MW natural gas combined cycle facility interconnecting at the Middlesex 138 kV substation in the New Jersey area. The request was received on October 26, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 480 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,145,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Deans-Halfmoon Bay 138.0 kV L	Jackass Flats-Paso Robles 138.0 kV	123.8	166	48.3
Middlesex-Halfmoon Bay Substation 138.0 kV	Jackass Flats-Paso Robles 138.0 kV	111.0	215	48.3
Jackass Flats-Middlesex 138.0 kV L	Halfmoon Bay-Middlesex 138.0 kV	103.1	228	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Deans-Halfmoon Bay 138.0 kV	Rebuild line with higher-capacity conductor	\$8,583,000	100% (\$8,583,000)
Middlesex-Halfmoon Bay 138.0 kV	Rebuild line with higher-capacity conductor	\$6,010,000	100% (\$6,010,000)
Jackass Flats-Middlesex 138.0 kV L	Reconductor line	\$552,000	100% (\$552,000)
	Total allocated to project		\$15,145,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.